1 Introduction to the C^2 Algebra

The theoretical constructs required for exact Fully Homomorphic Encryption (FHE) are becoming increasingly unconventional. Since the quaternion-based approach has proven unviable (the formal proof of which is omitted here), I am introducing a novel and highly unorthodox algebraic concept.

This new construction relies on embedding complex numbers within a complex number structure, tentatively denoted as C^2.

The core definition is straightforward: A C^2 element x is defined as a pair of complex numbers:

$$x = \big((a_1, a_2), (b_1, b_2)\big) \tag{1}$$

1.1 Algebraic Properties

Given that standard complex numbers satisfy all field (and thus ring) axioms, the C^2 numbers should theoretically preserve these ring axioms. Furthermore, unlike quaternions, C^2 elements are commutative.

Conjecture on Axioms: While commutativity and associativity hold trivially by definition, the distributivity axiom requires rigorous formal verification, though it is strongly conjectured to hold.

The identity elements in this structure are naturally defined as:

$$1' = \big((1, 0), (0, 0)\big) \tag{2}$$
$$0' = \big((0, 0), (0, 0)\big) \tag{3}$$

2 Cryptographic Significance of C^2

What makes this specific algebraic structure relevant for homomorphic encryption?

2.1 Orthogonal Idempotent Encoding (z/z')

Theoretically, it should be highly feasible to implement a z/z' encoding scheme within the C^2 algebra. This requires finding elements z and z' that act as orthogonal idempotents, satisfying the following system of equations:

$$z \cdot z' = 0' \tag{4}$$
$$z + z' = 1' \tag{5}$$
$$z^2 = z \tag{6}$$
$$(z')^2 = z' \tag{7}$$

From these properties, it follows that the norm evaluates to $|z| = |z'| = (0, 0)$. However, a crucial distinction in the C^2 space is that the norm of such an element is itself a complex number, not a real scalar.

I strongly conjecture that the C^2 structure provides a multiplicity of valid choices for the pair (z, z').

2.2 Fourier Transform Compatibility

Due to the preserved ring properties and commutativity, applying a Discrete Fourier Transform (DFT) over the C^2 algebra should remain mathematically well-defined.

3 Security Implications

The security of the proposed scheme relies on the combination of two factors:

1. **Multiplicity of Encodings:** The existence of multiple valid (z, z') pairs could introduce enough ambiguity to eliminate the scheme's reliance on the secrecy of the Prime Factorization (PFZ).
2. **Intractability of the DFT:** The Fourier Transform becomes practically intractable to invert algebraically if the exact origin parameters (the specific choice of z/z' and the permutation) remain unknown.

Therefore, this combination could potentially shield the underlying plaintext from the linear algebra attacks that defeated previous iterations.